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3.3.4 Householder QR factorization algorithm

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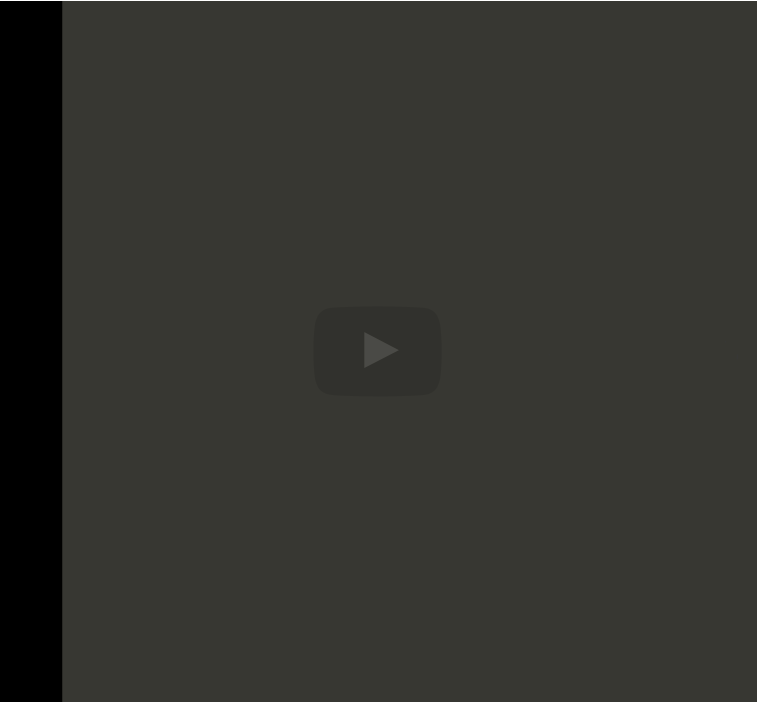
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u_2 Hermitian

transpose over tau_2.

You would need two

ones here and zeros

here. Okay? And when

you apply that to the

matrix -- well, first of

all we need to apply

that to A as

well -- so I'll just sort

of stop writing that.

And at the end of that,

you will

have introduced zeros

right here. And then we

compute a

Householder

transformation from

that. We apply it in

such a way that the

first three rows are

left alone, that

introduces a zero here.

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Reading assignment

1/1 point (graded)



Read Unit 3.3.4 of the notes. [\[LINK\]](#)

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Homework 3.3.4.1. 10	
Minor typo in instructions for HW 3.3.4.3. 2	

1/1 point (graded)
Show that

$$\left(\begin{array}{c|c} I & 0 \\ \hline 0 & I - \frac{1}{\tau_1} \begin{pmatrix} 1 \\ u_2 \end{pmatrix} \begin{pmatrix} 1 \\ u_2 \end{pmatrix}^H \end{array} \right) = \left(I - \frac{1}{\tau_1} \begin{pmatrix} 0 \\ 1 \\ u_2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ u_2 \end{pmatrix}^H \right).$$

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Solution

For a complete solution, see [the notes](#).

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Advanced Linear Algebra: Foundations to Frontiers

3.3.4
Householder QR factorization
algorithm Part 2

Robert van de Geijn
Maggie Myers

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So here I depicted sort of a typical state in the matrix, where in this case we've processed two columns, but those could represent an arbitrary number of columns, and the rest of the matrix has been updated according to the Householder transformations that were computed and applied in the before. So we can partition this into quadrants and say, "Okav this part

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Homework 3.3.4.2.

1/1 point (graded)
Given $A \in \mathbb{R}^{m \times n}$ show that the cost of the algorithm in

$[A, t] = \text{HQR_unb_var1}(A)$

$A \rightarrow \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \text{ and } t \rightarrow \begin{pmatrix} t_T \\ t_B \end{pmatrix}$

$A_{TL} \text{ is } 0 \times 0 \text{ and } t_T \text{ has 0 elements}$

$\text{while } n(A_{BR}) > 0$

$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c|c|c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right) \text{ and } \begin{pmatrix} t_T \\ t_B \end{pmatrix} \rightarrow \begin{pmatrix} t_0 \\ \tau_1 \\ t_2 \end{pmatrix}$

$\left[\begin{pmatrix} \alpha_{11} \\ a_{21} \end{pmatrix}, \tau_1 \right] := \left[\begin{pmatrix} \rho_{11} \\ u_{21} \end{pmatrix}, \tau_1 \right] = \text{Housev} \left(\begin{pmatrix} \alpha_{11} \\ a_{21} \end{pmatrix} \right)$

$\text{Update } \begin{pmatrix} a_{12}^T \\ A_{22} \end{pmatrix} := \left(I - \frac{1}{\tau_1} \begin{pmatrix} 1 \\ u_{21} \end{pmatrix} \begin{pmatrix} 1 & u_{21}^H \end{pmatrix} \right) \begin{pmatrix} a_{12}^T \\ A_{22} \end{pmatrix}$

via the steps

$w_{12}^T := (a_{12}^T + a_{21}^H A_{22}) / \tau_1$

$\begin{pmatrix} a_{12}^T \\ A_{22} \end{pmatrix} := \begin{pmatrix} a_{12}^T - w_{12}^T \\ A_{22} - a_{21} w_{12}^T \end{pmatrix}$

$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array}\right) \leftarrow \left(\begin{array}{cc|c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ \hline A_{20} & a_{21} & A_{22} \end{array}\right) \text{ and } \left(\begin{array}{c} t_T \\ \hline t_B \end{array}\right) \leftarrow \left(\begin{array}{c} t_0 \\ \hline t_1 \\ \hline t_2 \end{array}\right)$$

endwhile

is given by

$$C_{\text{HQR}}(m,n) \approx 2mn^2 - \frac{2}{3}n^3 \text{ flops.}$$

☒

done



Solution

For a complete solution, see [the notes](#).

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Homework 3.3.4.3

1/1 point (graded)
Implement the algorithm in

[A, t] = HQR_unb_var1(A)

$$A \rightarrow \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array}\right) \text{ and } t \rightarrow \left(\begin{array}{c} t_T \\ \hline t_B \end{array}\right)$$

A_{TL} is 0×0 and t_T has 0 elements

while $n(A_{BR}) > 0$

$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array}\right) \rightarrow \left(\begin{array}{cc|c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ \hline A_{20} & a_{21} & A_{22} \end{array}\right) \text{ and } \left(\begin{array}{c} t_T \\ \hline t_B \end{array}\right) \rightarrow \left(\begin{array}{c} t_0 \\ \hline t_1 \\ \hline t_2 \end{array}\right)$$

$$\left[\begin{pmatrix} \alpha_{11} \\ a_{21} \end{pmatrix}, \tau_1\right] := \left[\begin{pmatrix} \rho_{11} \\ u_{21} \end{pmatrix}, \tau_1\right] = \text{Housev}\left(\begin{pmatrix} \alpha_{11} \\ a_{21} \end{pmatrix}\right)$$

$$\text{Update } \begin{pmatrix} a_{12}^T \\ A_{22} \end{pmatrix} := \left(I - \frac{1}{\tau_1} \begin{pmatrix} 1 & \\ & u_{21}^H \end{pmatrix}\right) \begin{pmatrix} a_{12}^T \\ A_{22} \end{pmatrix}$$

via the steps

$$w_{12}^T := (a_{12}^T + a_{21}^H A_{22}) / \tau_1$$

$$\begin{pmatrix} a_{12}^T \\ A_{22} \end{pmatrix} := \begin{pmatrix} a_{12}^T - w_{12}^T \\ A_{22} - a_{21} w_{12}^T \end{pmatrix}$$

$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array}\right) \leftarrow \left(\begin{array}{cc|c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ \hline A_{20} & a_{21} & A_{22} \end{array}\right) \text{ and } \left(\begin{array}{c} t_T \\ \hline t_B \end{array}\right) \leftarrow \left(\begin{array}{c} t_0 \\ \hline t_1 \\ \hline t_2 \end{array}\right)$$

endwhile

as

function [A_out, t] = HQR(A)

by completing the code in [Assignments/Week03/matlab/HQR.m](#). Input is an $m \times n$ matrix A. Output is the matrix A_out with the Householder vectors below its diagonal and R in its upper triangular part. You may want to use [Assignments/Week03/matlab/test_HQR.m](#) to check your implementation.

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done



Solution

For a complete solution, see [the notes](#).

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